Mawlana Bhashani Science and Technology University

Department of ICT

Santosh, Tangail

Simple Mail Transfer Protocol (SMTP)

How the Internet sends, relays and delivers email

Presented By

MD. Samir Tahrian

ID: IT-24007

Dept. of ICT, MBSTU

Supervised By

Dr. Nazrul Islam

Associate Professor

Dept. of ICT, MBSTU

Course title: **Data Communication** | Course code: **ICT-3101**

Application-layer protocol | **Email transport**

Introduction and History of SMTP

SMTP is the standard application-layer protocol that moves email from a sender's mail client to the recipient's mail server.

TCP	25 · 587 · 465	RFC 5321	Push
Reliable transport	Well-known ports	Current core standard	Sender initiates transfer

History

1971 — First network email
Ray Tomlinson, ARPANET, uses the @ sign

1982 — RFC 821
SMTP standardised (RFC 788 in 1981 was first)

1995 — RFC 1869
ESMTP: EHLO and extensions

2008 — RFC 5321
Current SMTP standard

2018 — RFC 8314
TLS by default for submission

Email System Architecture



MUA: Mail User Agent

The client the person uses to write and read email.

MSA: Mail Submission Agent

Accepts new mail from clients, usually on port 587, after login.

MTA: Mail Transfer Agent

Relays mail between servers using SMTP (Postfix, Exim, Exchange).

MDA: Mail Delivery Agent

Places the final message into the user's mailbox.

How an Email Travels

1 Compose

The user writes the message in a mail client (MUA).

2 Submit

The client sends it to the outgoing server over SMTP, usually port 587, after login.

3 DNS lookup

The sending MTA asks DNS for the MX record of the recipient's domain.

4 Relay

It opens a TCP connection to the receiving MTA on port 25 and transfers the message.

5 Deliver

The MDA stores the message in the recipient's mailbox.

6 Retrieve

The recipient reads it using IMAP, POP3 or webmail.

DNS and MX records: a lower preference number means higher priority; backup MX servers accept mail if the primary is down; with no MX record the sender falls back to the domain's A/AAAA record.

Characteristics and Ports

Text-based: readable ASCII commands and replies ending in CRLF

Runs over TCP: reliable, ordered delivery

Push protocol: the sender pushes mail; it cannot pull

Client-server: sending side is the client, receiving side the server

Command-reply: every command gets a numeric reply code

Store and forward: servers queue mail and retry later

Port	Name	Purpose	Encryption
25	SMTP	Server-to-server relay (MTA to MTA)	Optional STARTTLS
587	Submission	Client to server submission (RFC 6409)	STARTTLS and login
465	SMTPS	Submission over implicit TLS (RFC 8314)	TLS from first byte
2525	Alternative	Unofficial fallback used by some providers	Usually STARTTLS

SMTP Commands and Reply Codes

Command	Purpose
EHLO	Greets the server and lists its extensions
MAIL FROM	Starts a transaction with the sender address
RCPT TO	Names one recipient (repeat for more)
DATA	Begins the message; ends with a lone dot line
RSET	Aborts the current transaction
VRFY	Checks whether a mailbox exists (often disabled)
STARTTLS	Upgrades the connection to TLS
AUTH	Authenticates the client (RFC 4954)
QUIT	Ends the session politely

2xx — **Success**
250 OK, 220 Ready, 221 Closing

3xx — **Intermediate**
354 Start mail input

4xx — **Temporary failure**
421 Unavailable, 450 Mailbox busy

5xx — **Permanent failure**
550 No such mailbox, 554 Failed

A Complete SMTP Session

```
S: 220 mx.example.com ESMTP ready
C: EHLO mail.sender.org
S: 250 mx.example.com Hello
C: MAIL FROM:<alice@sender.org>
S: 250 2.1.0 Sender OK
C: RCPT TO:<bob@example.com>
S: 250 2.1.5 Recipient OK
C: DATA
S: 354 Start mail input
C: From: Alice <alice@sender.org>
C: To: Bob <bob@example.com>
C: Subject: Hello
C:
C: Hi Bob, this is a test.
C: .
S: 250 2.0.0 Queued as 4F2A1
C: QUIT
S: 221 2.0.0 Closing connection
```

1. **Greeting**
   Server announces readiness (220); client identifies itself with EHLO.
2. **Envelope**
   MAIL FROM and RCPT TO name the sender and recipients.
3. **Data**
   Headers, a blank line, then the body, ended by a lone dot.
4. **Close**
   Server queues the message (250); QUIT ends the session.

Message Format and MIME

ENVELOPE (used by SMTP)

```
MAIL FROM:<a@sender.org>  RCPT TO:<b@example.com>
```

HEADER (RFC 5322)

```
Received: from mail.sender.org
From: Alice <alice@sender.org>
To: Bob <bob@example.com>
Subject: Meeting agenda
Date: Thu, 05 Nov 2026 11:09 +0600
Message-ID: <abc123@sender.org>
```

BODY

Text, HTML and attachments (MIME parts) after one blank line.

MIME (RFC 2045) lets 7-bit ASCII SMTP carry attachments, images and non-English text.

```
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary="X"
Content-Transfer-Encoding: base64
```

Content-Type	Used for
`text/plain, text/html`	Message text
`multipart/mixed`	Attachments
`multipart/alternative`	Plain + HTML
`application/pdf`	Documents

ESMTP Service Extensions

Extension	RFC	What it adds
STARTTLS	3207	Upgrade a plain connection to TLS encryption
AUTH	4954	Client authentication (PLAIN, LOGIN, OAuth)
SIZE	1870	Declare message size and server limit before sending
8BITMIME	6152	Send 8-bit data without extra encoding
PIPELINING	2920	Send several commands without waiting for each reply
DSN	3461	Delivery status notifications (success, delay, failure)
CHUNKING	3030	Transfer large messages in binary chunks (BDAT)
SMTPUTF8	6531	Internationalised addresses and headers (UTF-8)

SMTP vs POP3 vs IMAP

Feature	SMTP	POP3	IMAP
Purpose	Send and relay mail	Download mail	Access and sync mail
Direction	Push	Pull	Pull and synchronise
Ports	25, 587, 465	110, 995 (TLS)	143, 993 (TLS)
Mail stored	In server queues in transit	On the device; often removed from server	Kept on the server
Multiple devices	Not applicable	Limited	Excellent
Specification	RFC 5321	RFC 1939	RFC 9051 (IMAP4rev2)

Email Security: Threats and Protections

Threats

- Sender spoofing: From is not verified
- Spam and phishing
- Open relay abuse
- Eavesdropping (MITM) without TLS
- STARTTLS stripping (downgrade)
- Malware attachments and mail bombing

Protections

SPF	RFC 7208	Domain lists the IPs allowed to send its mail in DNS; receivers check the sending IP.
DKIM	RFC 6376	Sender signs the message with a private key; receivers verify with the public key in DNS.
DMARC	RFC 7489	Requires domain alignment with SPF or DKIM; policy is none, quarantine or reject.

Receiver checks: SPF, then DKIM, then DMARC policy, then inbox, spam or reject. Gmail and Yahoo require this for bulk senders (2024).

Also: TLS / STARTTLS, SMTP AUTH, MTA-STS and DANE

SMTP Software and Applications

Software / service	Highlights
Postfix	Open-source MTA by Wietse Venema (1998); secure and fast
Sendmail	Open-source MTA by Eric Allman (1983); very configurable
Exim	Open-source MTA from Cambridge (1995); Debian default
Microsoft Exchange	Enterprise mail and calendar server for Windows
Gmail SMTP	smtp.gmail.com on port 587 (STARTTLS) or 465 (TLS)
Amazon SES, SendGrid	Cloud services with API or SMTP interface for bulk mail

Real-world applications

- Transactional email: OTP, receipts, password resets
- Alerts: banking, security, monitoring
- Newsletters and marketing campaigns
- Institutional communication: universities, offices
- Automated reports and backups
- Web apps, IoT devices and printers via relay

Advantages, Limitations and Conclusion

Advantages

- Simple, text-based and easy to debug
- Universal: every provider can exchange mail
- Store-and-forward queues with retries
- Extensible through ESMTP

Limitations

- No built-in sender authentication
- Plain text unless TLS is used
- Push only: needs POP3 or IMAP to read mail
- Spam, abuse and size limits (e.g. 25 MB)

Conclusion: **after more than 40 years, SMTP remains the backbone of Internet email because it is simple, extensible and universal.**

Future: wider MTA-STS, DANE and DMARC enforcement, UTF-8 addresses and AI-based spam filtering.

Selected References

[1] J. Klensin, “Simple Mail Transfer Protocol,” RFC 5321, IETF, Oct. 2008.

[2] P. Resnick, “Internet Message Format,” RFC 5322, IETF, Oct. 2008.

[3] N. Freed and N. Borenstein, “MIME Part One: Format of Internet Message Bodies,” RFC 2045, Nov. 1996.

[4] P. Hoffman, “SMTP Service Extension for Secure SMTP over TLS,” RFC 3207, Feb. 2002.

[5] S. Kitterman, “Sender Policy Framework (SPF),” RFC 7208, Apr. 2014.

[6] D. Crocker, T. Hansen and M. Kucherawy, “DomainKeys Identified Mail (DKIM) Signatures,” RFC 6376, Sep. 2011.

[7] M. Kucherawy and E. Zwicky, “DMARC,” RFC 7489, Mar. 2015.

[8] K. Moore and C. Newman, “Cleartext Considered Obsolete: Use of TLS for Email Submission and Access,” RFC 8314, Jan. 2018.

[9] J. Kurose and K. Ross, Computer Networking: A Top-Down Approach. Pearson.

Thank you

Questions and Discussion